The documentation and process conversion measures necessary to comply with this revision shall be completed by 22 August 2001.

INCH-POUND

MIL-PRF-19500/444E
24 May 2001
SUPERSEDING
MIL-PRF-19500/444D
18 June 1999

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING
TYPES 1N5711-1, 1N5711UR-1, 1N5712-1, 1N5712UR-1,
1N6857-1, 1N6857UR-1, 1N6858-1 AND 1N6858UR-1
JAN, JANTX, JANTXV, JANJ, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for Schottky barrier diodes. Five levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1, similar to DO-35, figure 2, UR package, DO-213AA, and figure 3, JANHC and JANKC.

1.3 Maximum ratings.

Type	V_{RWM}	I_{O1}	I_{O2} (1)	$R_{\theta JL}$ (2)	T_{STG} and T_J	ESDS class
	V(pk)	mA dc	mA dc	°C/W		
1N5711-1	50	33 (3)	5	250	-65°C to +150°C	1
1N5711UR-1	50	33 (4)	5	100		1
1N6858-1	50	75 (5)	10	250		2
1N6858UR-1	50	75 (6)	10	100		2
1N5712-1	16	75 (5)	20	250		1
1N5712UR-1	16	75 (6)	20	100		1
1N6857-1	16	150 (7)	50	250		2
1N6857UR-1	16	150 (8)	50	100		2

(1) Maximum I_O rating to ensure τ_{CL} compliance (< 100 ps).
(2) Lead length = .375 inch (9.52 mm) except for surface mount.
(3) At T_L = +130°C and L = .375 inch, derate I_O to 0 at +150°C.
(4) At T_{EC} = +140°C, derate I_O to 0 at +150°C.
(5) At T_L = +110°C and L = .375 inch, derate I_O to 0 at +150°C
(6) At T_{EC} = +130°C, derate I_O to 0 at +150°C.
(7) At T_L = +70°C and L = .375 inch, derate I_O to 0 at +150°C
(8) At T_{EC} = +110°C, derate I_O to 0 at +150°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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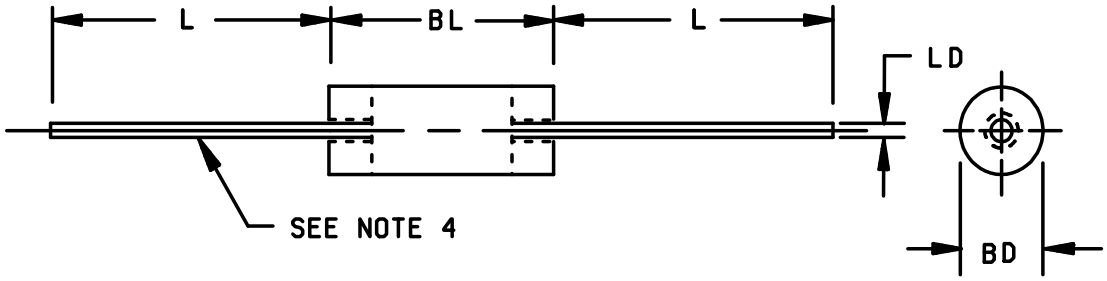
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1.4 Primary electrical characteristics.

Type	$V_{(BR)1}$ I_R = 10 µA dc (min)	V_{F1} I_F = 1 mA dc (max)	V_{F2} I_F = 15 mA dc (max)	I_{R1} V_R = 50 V dc (max)	C V_R = 0, f = 1 MHz V_{sig} = 50 m V(pk) (max)	τ_{CL} (max)		$Z_{\theta JX}$ (max)
	V dc	V dc	V dc	nA dc	pF	ps	mA	°C/W
1N5711-1, 1N5711UR-1	70	0.410	1.0	200	2.0	100	5 mA	40
1N6858-1, 1N6858UR-1	70	0.360	0.65	200	4.5	100	10 mA	40

Type	$V_{(BR)1}$ I_R = 10 µA dc (min)	V_{F1} I_F = 1 mA dc (max)	V_{F2} I_F = 35 mA dc (max)	I_{R1} V_R = 16 V dc (max)	C V_R = 0, f = 1 MHz V_{sig} = 50 mv(pk) (max)	τ_{CL} (max)		$Z_{\theta JX}$ (max)
	V dc	V dc	V dc	nA dc	pF	ps	mA	°C/W
1N5712-1, 1N5712UR-1	20	0.410	1.0	150	2.0	100	20 mA	40
1N6857-1, 1N6857UR-1	20	0.350	0.75	150	4.5	100	50 mA	40

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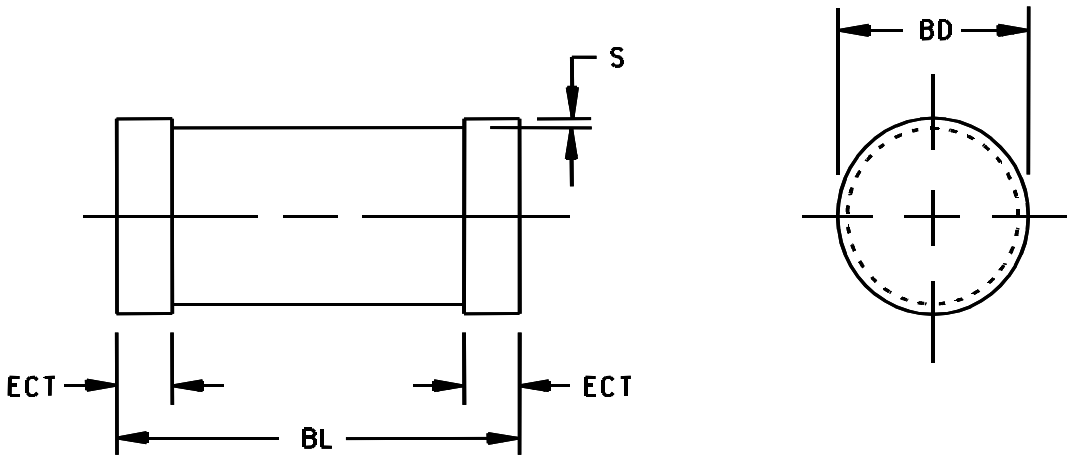


Dimensions					
	Inches		Millimeters		
Ltr	Min	Max	Min	Max	Notes
BL	.150	.170	3.81	4.32	3
BD	.068	.076	1.73	1.93	3, 4
L	1.000	1.500	25.40	38.10	
LD	.014	.022	0.36	0.56	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions BL and LD include all components of the diode periphery except the sections of the leads over which the diameter is controlled.
4. Dimension BD shall be measured at the largest diameter.
5. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for DO-35.

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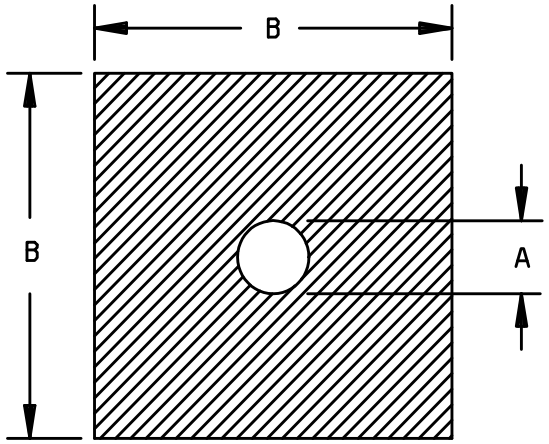


Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
BD	.063	.067	1.60	1.70	
ECT	.016	.022	0.41	0.55	
BL	.130	.146	3.30	3.71	
S	.001 Min		0.03 Min		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions for DO-213AA.

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BACKSIDE IS CATHODE

1N5711 Dimensions					
	Inches		Millimeters		
Ltr	Min	Max	Min	Max	Notes
A	.004	.005	.102	.127	
B	.0130	.0170	.330	.430	

1N5712 Dimensions					
	Inches		Millimeters		
Ltr	Min	Max	Min	Max	Notes
A	.0029	.0035	.074	.089	
B	.0130	.0170	.330	.430	

1N6857, 1N6858 Dimensions					
	Inches		Millimeters		
Ltr	Min	Max	Min	Max	Notes
A	.0055	.0065	.140	.165	
B	.0130	.0170	.330	.430	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Element evaluation accomplished utilizing TO-5 package.
4. The physical characteristics of the die are:
   Metallization:
   Top (anode): Al.
   Back (cathode): Au.
   Al thickness: 25,000 Å minimum.
   Gold thickness: 4,000 Å minimum.
   Chip thickness: .010 inch (0.25 mm) ±.002 inch (0.05 mm).

FIGURE 3. Physical dimensions, JANHCA and JANKCA die.

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2. APPLICABLE DOCUMENTS

2.1 General.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

DEFENSE ELECTRONICS SUPPLY CENTER

C68001 - Test Fixture for Effective Minority Carrier Lifetime.

(Copies of DESC drawing C68001 are available from Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.4).

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3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

EC	End cap.
UR	Surface mount case outline, round end cap.
τ_{CL}	Effective carrier lifetime.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500 and figures 1 (similar to DO-35), 2 (DO-213AA), and 3 (die) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. All devices shalle be merallurgically bonded double plug construction in accordance with the requirements of category I or II (see MIL-PRF-19500).

3.4.3 UR devices. "UR" devices shall be marked with a cathode band only. Initial container package marking will be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I.

3.6 Electrical test requirements. The electrical test requirements shall be table I, subgroup 2.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.1.1 Sampling inspection. Sampling inspection shall be in accordance with MIL-PRF-19500.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and table II herein.

4.2.2 JANHC and JANKC devices. Qualification shall be in accordance with MIL-PRF-19500.

4.2.3 JANJ devices. For JANJ level, 3.3.1 through 3.3.1.3 of MIL-PRF-19500 shall apply, except as modified herein. Supplier imposed requirements as well as alternate screens, procedures, and/or controls shall be documented in the QM plan and must be submitted to the qualifying activity for approval. When alternate screens, procedures, and/or controls are used, in lieu of the JANJ screens herein, equivalency shall be proven and documented in the QM plan. Radiation characterization may be submitted in the QM plan at the option of the manufacturer, however, 3.3.1.1 of MIL-PRF-19500 is not required. Die lot control and rework shall be in accordance with 3.13 and D3.13.2.1 of MIL-PRF-19500 for JANS level. Lot formation and conformance inspection requirements for JANJ shall be those used for JANTXV devices as a minimum.

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4.2.4 JANJ qualification. For JANJ qualification, 4.4.2.1 herein shall be performed as required by the qualifying activity.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS Level	JANJ Level
1a	Required	Not required
1b	Required	Required
2	Not required	Not required
3a 3b (1) 3c	Required Not applicable Required (see 4.3.3)	Required Not applicable Required (see 4.3.3)
4, 5, 6 and 7a	Not applicable	Not applicable
7b	Required	Required
8	Required	Not required
9 and 10	Not applicable	Not applicable
11	Required I_{R1} and V_{F1}	Required I_{R1} and V_{F1}
12	Required, see 4.3.1	Required, see 4.3.1
(2) 13	Required. Subgroup 2 and 3 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or 25 nA whichever is greater; $\Delta V_{FM1} \leq \pm$ 20 mV dc.	Required. Subgroup 2 and 3 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or 25 nA whichever is greater; $\Delta V_{FM1} \leq \pm$ 20 mV dc.
14a 14b	Not applicable Optional (3)	Not applicable Optional (3)
15	Required	Required Attributes data only, film or non-film techniques may be utilized.
16	Required	Required
17	Not applicable	Required, subgroup 2 of table I herein.

See notes at end of table.

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4.3 Screening (JAN, JANTX, JANTXV, JANJ, and JANS levels only)- continued.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANTXV and JANTX Level	JAN level
1a	Not required	Not required
1b	Required (JANTXV only)	Not required
2	Not required	Not required
3a	Required	Required in accordance with MIL-PRF-19500, JANTX level
3b	Not applicable	Not applicable
(1) 3c	Required (see 4.3.1)	Required (see 4.3.1)
4, 5, 6 and 7a	Not applicable	Not applicable
7b	Required	Not required
8	Not required	Not required
9 and 10	Not applicable	Not applicable
11	Required. I_{R1} and V_{F1}	Not applicable
12	Required. See 4.3.1, t = 48 hours	Not applicable
(2) 13	Subgroup 2 of table I herein; $\Delta I_{R1} \leq 100$ percent of initial reading or 50 nA whichever is greater; $\Delta V_{F1} \leq \pm 40$ mV dc.	Not applicable
14a	Not applicable	Not applicable
14b	Optional (3)	(Not required)
15 and 16	Not required	Not required
17	Not applicable	Not applicable

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with table IV of MIL-PRF-19500, screen 3 prior to this thermal test.
2/ Test within 24 hours after removal from test.
3/ In accordance with MIL-PRF-19500.

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4.3.1 Burn-in conditions. Burn-in conditions are as follows: High temperature reverse bias (HTRB), method 1038 of MIL-STD-750, test condition A, V_R = 40 V; T_A = +150°C (min.) 1N5711, 1N6858. V_R = 12.8V TA = +150°C (min) 1N5712, 1N6857.

4.3.2 Screening (JANHC or JANKC). Screening of JANC die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100-percent probed in accordance with group A, subgroup 2 (excluding thermal impedance).

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750 to identify and remove atypical devices.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. I_M measurement current .....1 mA to 10 mA.

b. I_H forward heating current .....50 mA to 200 mA.

c. t_H heating time .....10 ms.

d. t_{MD} measurement delay time.....70 µs maximum.

4.4.2 Group B inspection . Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (for JANS) and table VIb (for JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	I_{O1} at rated V_{RWM}; f = 60 Hz; T_A = room ambient as defined in 4.5 of MIL-STD-750; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles; (see 4.5.1).
B5	1038	High temperature reverse bias (HTRB), in accordance with 4.3.1.
B6	3101 or 4081	$R_{\theta JL}$ = 250°C/W .375 inch (9.52 mm) lead length (non-surface mount). $R_{\theta JEC}$ = 100°C/W (UR suffix devices).

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, JANTXV, and JANJ) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1038	High temperature reverse bias (HTRB), in accordance with 4.3.1.
B4	See 4.5.4.	
B5	Not applicable.	
B6	1032	T_A = +150°C.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	(Not applicable to surface mount devices); lead fatigue conditions: Test condition E; .062 inch (1.57 mm) lead restriction from case. Tension conditions: Test condition A; 4 pounds, 15 seconds.
C6	1038	High temperature reverse bias (HTRB), in accordance with 4.3.1.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180 degrees nor less than 150 degrees.

4.5.2 Effective carrier lifetime (τ_{CL}). This test shall be measured with the device installed in the test fixture, DESC drawing C68001 using the test setup shown on figure 4 herein. Adjust the signal generator to 54 MHz and maximum output in order to obtain a sufficient trigger signal for the oscilloscope. Adjust the sensitivity of the oscilloscope to 5 mV/cm (1N5711), 10mV/cm (1N6858), 20mV/cm (1N5712), and 50mV/cm (1N6857). With the device in the test fixture, adjust the output of the amplifier until the peak amplitude of the forward current is 5 cm as seen on the oscilloscope [5 mA (1N5711), 10mA (1N6858), 20mA (1N5712) and 50mA (1N6857)].

Change the sensitivity of scope to 2 mV/cm. Under these conditions, the effective carrier lifetime is related to the amplitude designated as "τ_{CL} " shown on figure 5. This amplitude has the calibration of 200 ps/cm (1N5711), 100ps/cm (1N6858), 50ps/cm (1N5712), and 20ps/cm (1N6857).

4.5.3 Thermal resistance. Thermal resistance measurement shall be in accordance with method 3101 or 4081 of MIL-STD-750. Forced moving air or draft shall not be permitted across the device during test. The maximum limit for $R_{\theta JL}$ under these test conditions shall be as shown in group E. The following conditions shall apply when using method 3101:

a. I_M ..... 1 mA to 10 mA.

b. I_H ..... 25 mA to 100 mA.

c. t_H ..... 25 seconds minimum.

d. t_{MD} ..... 70 µs maximum.

4.5.3.1 Temperature reference point, leaded devices: LS = lead spacing = .375 inch (9.52 mm) as defined on figure 6.

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4.5.3.2 Temperature reference point, surface mount. The temperature reference point shall be the hottest point on the external package surface. As an alternate, the temperature reference point may be the temperature of a liquid stream used to cool the device during testing.

4.5.4 Decap internal visual scribe and break. Scratch glass at cavity area with diamond scribe. Carefully snap open. Using 30X magnification, examine the area where die (or bonding material) are in contact with the plugs, verify metallurgical bonding area.

TABLE I. Group A inspection .

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2 1N5711						
Forward voltage	4011	I_F = 1 mA dc	V_{F1}		0.410	V dc
Forward voltage	4011	I_F = 15 mA dc	V_{F2}		1.0	V dc
Reverse current	4016	DC method; V_R = 50 V dc	I_{R1}		200	nA dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)1}$	70		V dc
Thermal impedance	3101	see 4.4.1	$Z_{\theta JX}$		40	°C/W
Subgroup 2 1N5712						
Forward voltage	4011	I_F = 1 mA dc	V_{F1}		0.410	V dc
Forward voltage	4011	I_F = 35 mA dc	V_{F2}		1.0	V dc
Reverse current	4016	DC method; V_R = 16 V dc	I_{R1}		150	nA dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)1}$	20		V dc
Thermal impedance	3101	see 4.4.1	$Z_{\theta JX}$		40	°C/W
Subgroup 2 1N6857						
Forward voltage	4011	I_F = 1 mA dc	V_{F1}		0.350	V dc
Forward voltage	4011	I_F = 35 mA dc	V_{F2}		0.75	V dc
Reverse current	4016	DC method; V_R = 16 V dc	I_{R1}		150	nA dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)1}$	20		V dc
Thermal impedance	3101	see 4.4.1	$Z_{\theta JX}$		40	°C/W
Subgroup 2 1N6858						
Forward voltage	4011	I_F = 1 mA dc	V_{F1}		0.360	V dc
Forward voltage	4011	I_F = 15 mA dc	V_{F2}		0.65	V dc
Reverse current	4016	DC method; V_R = 50 V dc	I_{R1}		200	nA dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)1}$	70		V dc
Thermal impedance	3101	see 4.4.1	$Z_{\theta JX}$		40	°C/W

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 1N5711						
High temperature operation	4011	T_A = +150°C				
Reverse current	4016	DC method; V_R = 50 V dc	I_{R2}		200	µA dc
Low temperature operation	4011	T_A = -55°C				
Forward voltage	4011	I_F = 1 mA dc	V_{F3}		0.55	V dc
Forward voltage	4011	I_F = 15 mA dc	V_{F4}		1.0	V dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)2}$	70		V dc
Subgroup 3 1N5712						
High temperature operation	4011	T_A = +150°C				
Reverse current	4016	DC method; V_R = 16 V dc	I_{R2}		150	µA dc
Low temperature operation	4011	T_A = -55°C				
Forward voltage	4011	I_F = 1 mA dc	V_{F3}		0.55	V dc
Forward voltage	4011	I_F = 35 mA dc	V_{F4}		1.0	V dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)2}$	20		V dc
Subgroup 3 1N6857						
High temperature operation	4011	T_A = +150°C				
Reverse current	4016	DC method; V_R = 16 V dc	I_{R2}		150	µA dc
Low temperature operation	4011	T_A = -55°C				
Forward voltage	4011	I_F = 1 mA dc	V_{F3}		0.55	V dc
Forward voltage	4011	I_F = 35 mA dc	V_{F4}		1.0	V dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)2}$	20		V dc
Subgroup 3 1N6858						
High temperature operation	4011	T_A = +150°C				
Reverse current	4016	DC method; V_R = 50 V dc	I_{R2}		200	µA dc
Low temperature operation	4011	T_A = -55°C				
Forward voltage	4011	I_F = 1 mA dc	V_{F3}		0.55	V dc
Forward voltage	4011	I_F = 15 mA dc	V_{F4}		1.0	V dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)2}$	70		V dc
Subgroup 4 1N5711						
Capacitance	4001	V_r = 0V dc; f = 1 MHz; V_{sig} = 50 mV(pk) maximum	C			
1N5711, 1N5712, 1N6857, 1N6858				2.0 4.5		pF pF
Effective carrier 2/ Lifetime (see 4.5.2)		(see DESC drawing C68001)	τ_{CL}		100	
Subgroups 5, 6, and 7 Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ Effective carrier lifetime needs to be performed only one time per wafer lot.

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TABLE II. Group E inspection (all quality levels) - for qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			22 devices c = 0
Temperature cycling (air to air)	1051	500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See group A, subgroup 2 (V_{F1} and I_{R1})	
Subgroup 2			22 devices c = 0
Steady-state dc blocking life	1038	See 4.3.1, 1,000 hrs	
Electrical measurements		See group A, subgroup 2 (V_{F1} and I_{R1})	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance surface mount (UR)	3101 or 4081	$R_{\theta JEC}$ = 100°C/W (maximum) at zero lead length. +25°C ≤ T_R ≤ +35°C, (see 4.5.3) at T_H ≥ 25 s	
Thermal resistance leaded	3101 or 4081	$R_{\theta JEC}$ = 250°C/W (maximum) at .375 inches (9.52 mm) lead length. +25°C ≤ T_R ≤ +35°C, (see 4.5.3) at T_H ≥ 25 s in still air.	
Subgroup 6			3 devices c = 0
ESD testing	1020		

1/ For sampling plan, see MIL-PRF-19500.

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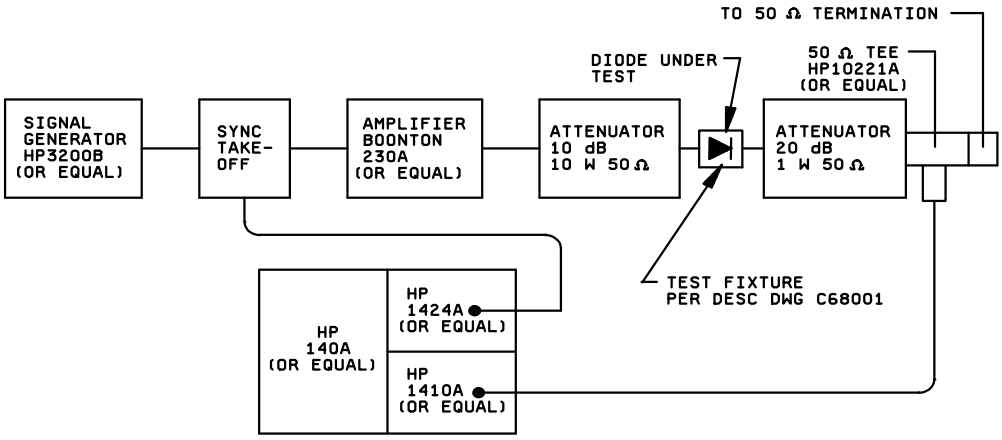


FIGURE 4. Test setup for diode effective lifetime measurement.

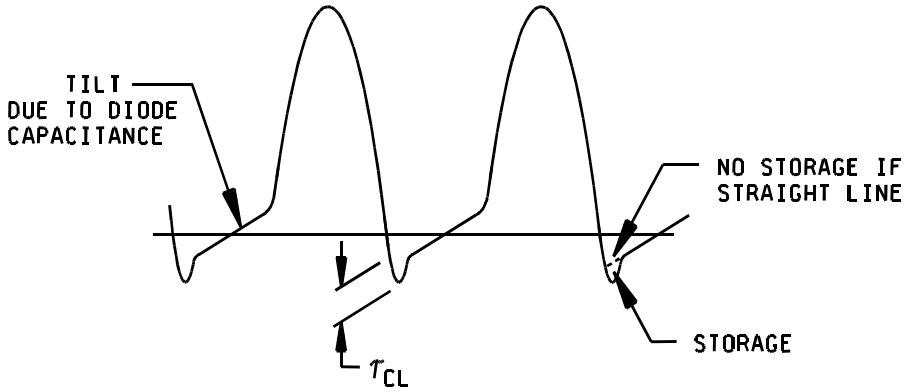


FIGURE 5. Oscilloscope display in diode effective lifetime measurement.

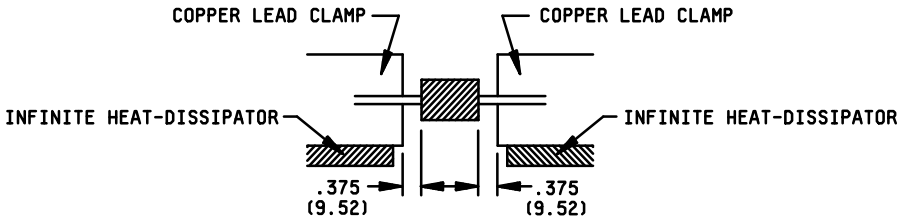


FIGURE 6. Mounting conditions.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DoDISS to be cited in the solicitation.

c. Lead finish as specified (see 3.4.1).

d. Product assurance level, type designator, and for die acquisition the JANHC and JANKC identification (see figure 3 and 6.3).

e. Packaging. requirements (see 5.1).

6.3 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5711) will be identified on the QML.

JANC ordering information	
PIN	Manufacturer
	55801
1N5711-1	JANHCA1N5711
1N5711-1	JANKCA1N5711
1N5712-1	JANHCA1N5712
1N5712-1	JANKCA1N5712
1N6857-1	JANHCA1N6857
1N6857-1	JANKCA1N6857
1N6858-1	JANHCA1N6858
1N6858-1	JANKCA1N6858

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - NW
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2434)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/444E	2. DOCUMENT DATE 24 May 2001

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING TYPES 1N5711-1, 1N5711UR-1, 1N5712-1, 1N5712UR-1, 1N6857-1, 1N6857UR-1, 1N6858-1 AND 1N6858UR-1 JAN, JANTX, JANTXV, JANJ, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (PLEASE PRINT)	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan.barone@dscc.dla.mil
c. ADDRESS *(Include Zip Code)* Defense Supply Center Columbus ATTN: DSCC-VAC P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703)767-6888 DSN 427-6888

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